ECON 315: International Economics
Problem Set #3

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Problem 1

Restatement: The domestic demand and supply curves for towels in Guatemala are

$$P = 66 - 2Q \quad \text{(demand)}, \qquad P = 6 + Q \quad \text{(supply)},$$

where P is in pesos and Q is in towels. Assume a one-to-one peso–dollar exchange rate throughout.

(a) Autarky price and quantity

Solution: Set $Q_d = Q_s$:

$$66 - 2Q = 6 + Q \implies 60 = 3Q \implies Q_A = 20.$$

Substituting back, $P_A = 6 + 20 = \mathbf{26}$ pesos at $Q_A = \mathbf{20}$ towels.

(b) Free trade at the world price $P_W = \$20$

Solution: Since $P_W = 20 < P_A = 26$, Guatemala is a net importer.

- Domestic production: $20 = 6 + Q_s \implies Q_s = \mathbf{14}$.
- Domestic consumption: $20 = 66 - 2Q_d \implies Q_d = \mathbf{23}$.
- Imports: $Q_d - Q_s = 23 - 14 = \mathbf{9}$ towels.

(c) A 10% ad valorem tariff

Solution: Guatemala is small, so the tariff passes through fully to the domestic price: $P_T = 20 \times 1.10 = \$\mathbf{22}$.

- Domestic production: $22 = 6 + Q'_s \implies Q'_s = \mathbf{16}$.
- Domestic consumption: $22 = 66 - 2Q'_d \implies Q'_d = \mathbf{22}$.
- Imports: $Q'_d - Q'_s = 22 - 16 = \mathbf{6}$ towels.

The tariff raises domestic production by 2 towels, contracts consumption by 1 towel, and cuts imports by 3 towels relative to free trade.

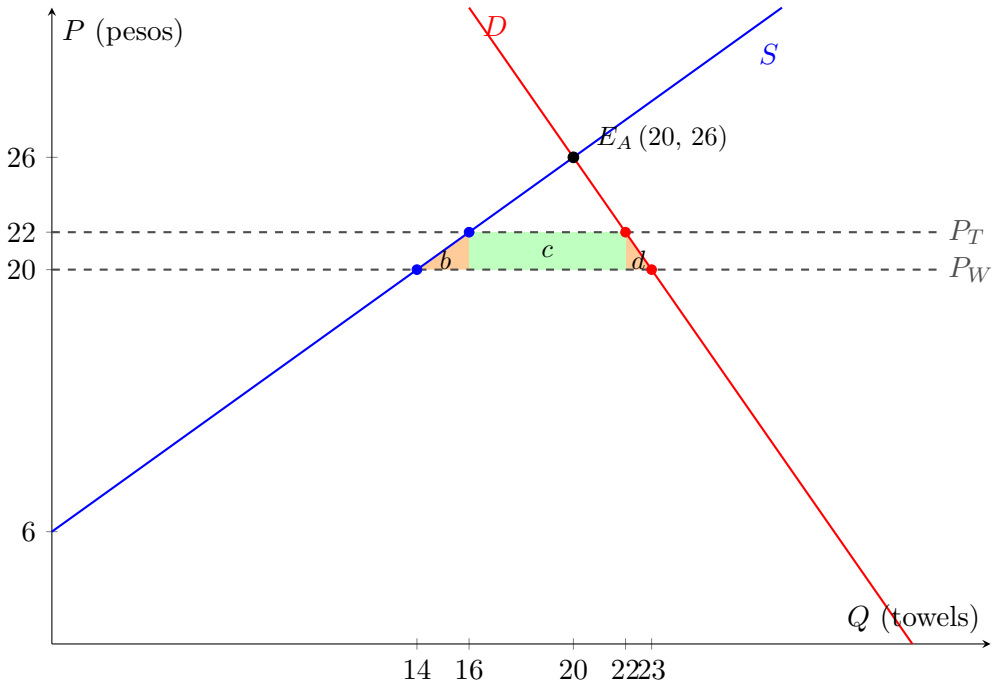

Figure 1: Tariff effects in Guatemala. Free-trade imports $= 23-14 = 9$; tariff imports $= 22-16 = 6$. Triangle b (area $= 2$) is the production-distortion loss; triangle d (area $= 1$) is the consumption-distortion loss; rectangle c (area $= 12$) is tariff revenue.

(d) Deadweight cost of the tariff (in pesos)

Solution: The deadweight loss (DWL) consists of two triangles in the diagram above:

$$\underbrace{\text{Production distortion } (b)\text{:}}_{\text{area}=\frac{1}{2}(\Delta Q_s)(\Delta P)} = \tfrac{1}{2}(16-14)(22-20) = 2 \text{ pesos},$$

$$\underbrace{\text{Consumption distortion } (d)\text{:}}_{\text{area}=\frac{1}{2}(\Delta Q_d)(\Delta P)} = \tfrac{1}{2}(23-22)(22-20) = 1 \text{ peso}.$$

$$\boxed{\text{Total DWL} = b + d = \mathbf{3} \text{ pesos.}}$$

Consistency check: Loss of consumer surplus $= 45$ pesos; gain of producer surplus $= 30$ pesos; tariff revenue $= 12$ pesos. Net welfare effect $= 45 - 30 - 12 = 3$ pesos, matching the DWL exactly.

(e) Tariff revenue

Solution: Revenue is the per-unit tariff times the number of units imported under the tariff:

$$\text{Revenue} = (P_T - P_W) \times \text{Imports}_T = \$2 \times 6 = \mathbf{12} \text{ pesos}.$$

This corresponds to the shaded green rectangle (area c) in the figure.

(f) Deadweight cost with less-elastic demand and/or supply

Solution: The DWL would be **smaller**. The two triangles have area $\frac{1}{2}(\Delta Q)(\Delta P)$; while the price wedge ΔP is fixed by the tariff, ΔQ is governed by the slope of the relevant curve. A less-elastic (steeper) curve means quantity responds less to the price change, which shrinks the base of the triangle. In the limit of perfect inelasticity, producers and consumers cannot reallocate at all in response to the tariff—no distortion occurs and DWL collapses to zero. This is the same logic underlying the Ramsey principle that distortionary taxes should fall on the most inelastic margins; the tariff is efficient (in the narrow sense of minimizing DWL) precisely when it has little behavioral bite.

(g) Large-country case

Solution: If Guatemala were large, the tariff would reduce its import demand enough to move the world price. Because foreign exporters face a smaller market, they must accept a lower price to clear the residual quantity; the world price of towels, P_W, **falls**. The domestic price rises by less than the full tariff, and the wedge between domestic and world prices is partially absorbed by foreign exporters—a terms-of-trade transfer *to* Guatemala.

Guatemala's terms of trade (export prices over import prices) therefore **improve**, because towels are an import and the import price has fallen. Whether this improvement translates into a net welfare gain depends on the balance between two effects:

- *Terms-of-trade gain* (rectangle e in the standard large-country diagram): the government collects revenue not just from domestic consumers but also from foreign exporters via the lower world price.
- *Volume-of-trade loss* (still the triangles $b + d$): trade contracts, forgoing mutually beneficial exchange.

The net effect is positive only if (i) foreign export supply to Guatemala is sufficiently inelastic, so the terms-of-trade gain is large; (ii) the tariff rate is at or below the *optimum tariff* rate $t^* = 1/\varepsilon_f$, where ε_f is the elasticity of the foreign offer curve; and (iii) trade partners do not retaliate. If partners retaliate with optimum tariffs of their own, both nations typically end up on lower trade indifference curves than under free trade—one of the clearest cases where a Nash equilibrium is strictly dominated by the cooperative outcome.

Problem 2

Restatement: Analyze transportation costs between two countries using supply and demand diagrams. The free-trade (no-transport-cost) price in both countries is \$6 per unit of wheat; autarky prices are \$3 in country A and \$9 in country B. A transportation cost of \$1 is split equally. What does equal sharing assume about elasticities? If country A has a more inelastic set of curves than B, where does the burden of the \$1 fall more heavily, and why?

Setup and the wedge created by transport costs

Because $P_A^{\text{autarky}} = \$3 < P_B^{\text{autarky}} = \$9$, country A has comparative advantage in wheat and exports to country B. Under free trade *without* transport cost, the world price settles at \$6 in both countries.

A transport cost of \$1 drives a wedge between the price received by A's producers and the price paid by B's consumers:

$$P_B - P_A = \$1, \qquad \text{and with equal sharing,} \qquad P_A = \$5.50, \quad P_B = \$6.50.$$

To make the geometry concrete, I use symmetric linear curves that generate the stated autarky prices:

$$\text{Country A: } S_A : P = \tfrac{Q}{4}, \quad D_A : P = 6 - \tfrac{Q}{4}; \qquad \text{Country B: } S_B : P = 6 + \tfrac{Q}{4}, \quad D_B : P = 12 - \tfrac{Q}{4}.$$

These curves intersect at $(Q, P) = (12, 3)$ for A and $(12, 9)$ for B—matching the stated autarky prices.

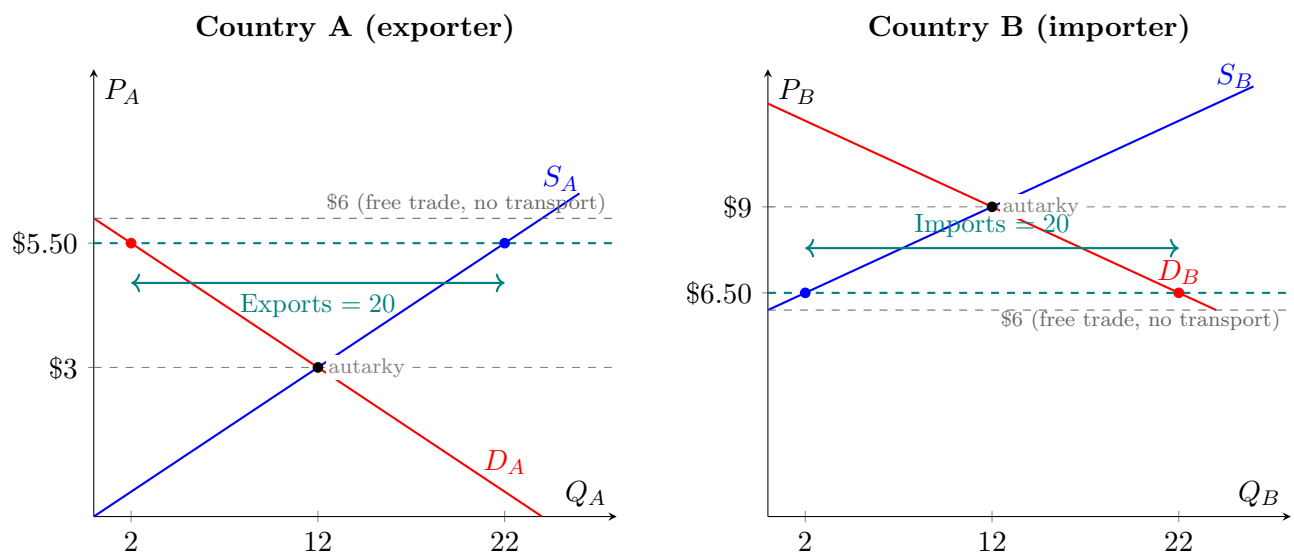

Figure 2: With the \$1 transport cost split evenly, A's producers receive \$5.50 while B's consumers pay \$6.50. At these prices, A exports 20 units ($Q_s = 22$, $Q_d = 2$) and B imports 20 units ($Q_d = 22$, $Q_s = 2$)—the market clears. Autarky prices (\$3 and \$9) are marked for reference.

What does equal sharing assume about elasticities?

Solution: Equal sharing is a special—and knife-edge—case. Formally, if η_A denotes the elasticity of A's *excess supply* (its export supply curve) and η_B denotes the elasticity of B's *excess demand* (its import demand curve), the incidence of the transport cost splits as

$$\text{Share borne by A} = \frac{\eta_B}{\eta_A + \eta_B}, \qquad \text{Share borne by B} = \frac{\eta_A}{\eta_A + \eta_B}.$$

- Equal sharing requires $\eta_A = \eta_B$.
- In other words, A's excess-supply curve and B's excess-demand curve must be *equally elastic* at the trading equilibrium.
- The symmetric linear curves drawn above are constructed to satisfy this condition exactly.

Incidence when A has more-inelastic curves than B

Solution: If A's supply and demand are more inelastic than B's, then $\eta_A < \eta_B$. Substituting into the incidence formula:

$$\text{Share borne by A} = \frac{\eta_B}{\eta_A + \eta_B} > \tfrac{1}{2}.$$

The burden falls more heavily on country A. The logic is the same as in tax incidence generally: the less-responsive side of the market bears more of the wedge. A's producers cannot easily divert wheat to other uses if prices fall, and A's consumers cannot easily substitute away from wheat if prices rise—so when the transport cost drives prices apart, A must absorb most of the adjustment. B, by contrast, has elastic room to reduce import demand if its import price edges up, forcing A to accept a larger price decline. In the limit where A's curves approach perfect inelasticity, A bears the entire \$1; in the opposite limit, B bears it all. This is a structural result, not a policy choice: "equal sharing" only describes the equilibrium when the two economies are equally flexible.

Problem 3

Restatement: The effective rate of protection (ERP) is

$$g = \frac{t_f - a\,t_i}{1 - a},$$

where t_f is the nominal tariff on the final good, t_i is the nominal tariff on imported inputs, and a is the ratio of the imported input cost to the final good's free-trade price.

(a) Show that if $a = 0$, then $g = t_f$

Solution: Substituting $a = 0$ into the formula:

$$g = \frac{t_f - (0)\,t_i}{1 - 0} = \frac{t_f}{1} = t_f. \qquad \blacksquare$$

Interpretation: When domestic production uses no imported inputs, the entire product price *is* domestic value added. The nominal tariff on the final good is therefore equivalent to the tariff on domestic value added, so nominal and effective protection coincide. This collapse is the benchmark against which all effective-protection calculations are compared: any divergence between g and t_f arises solely from the input-tariff structure.

(b) Compute g for $t_f = 0.5$, $t_i = 0.2$, $a = 0.4$

Solution:

$$g = \frac{0.5 - (0.4)(0.2)}{1 - 0.4} = \frac{0.5 - 0.08}{0.6} = \frac{0.42}{0.6} = \mathbf{0.70} = \mathbf{70}\%.$$

Interpretation: Although the nominal tariff on the final good is only 50%, domestic producers enjoy effective protection of 70% on their value-added activities. The reason is tariff escalation: the input tariff ($t_i = 20\%$) is well below the output tariff ($t_f = 50\%$), so the tariff structure taxes finished goods more than the inputs used to make them. Each peso of domestic value added is therefore sheltered from foreign competition by more than the headline 50% rate suggests. This pattern—high tariffs on finished consumer goods, low tariffs on intermediate inputs—is precisely the structure developed countries use to protect domestic manufacturing while keeping raw material costs competitive.

(c) Compute g for $a = 0.5$, $t_f = 0.4$, and $t_i = 0.8, 1.0$

Solution:

Case $t_i = 0.8$:

$$g = \frac{0.4 - (0.5)(0.8)}{1 - 0.5} = \frac{0.4 - 0.4}{0.5} = \mathbf{0}.$$

Effective protection is *zero*—the input tariff exactly offsets the output tariff, and domestic value added is no more protected than it would be under free trade. The tariff structure is neutral with respect to the industry's value-added margin; the final-good tariff raises output prices, but the input tariff raises input costs by an equal and opposite amount.

Case $t_i = 1.0$:

$$g = \frac{0.4 - (0.5)(1.0)}{1 - 0.5} = \frac{-0.1}{0.5} = \mathbf{-0.20} = \mathbf{-20}\%.$$

Effective protection is *negative*. The 100% tariff on imported inputs raises production costs by more than the 40% output tariff raises revenues, so domestic value added is actually *lower* than it would be under free trade. This is the pathological outcome of a poorly-designed tariff structure: nominally "protectionist" policies that, in practice, make it harder for domestic producers to compete. Developing countries that impose heavy tariffs on capital goods while protecting consumer goods often discover—belatedly—that they have engineered negative effective protection in their own intermediate industries.

(d) Main drawbacks of the effective-protection concept

Solution: Three drawbacks are worth highlighting:

- *Fixed input-output coefficients (Leontief assumption).* The ERP formula assumes a is a technological constant, but producers substitute between imported and domestic inputs when relative prices change. The formula captures a first-order tariff effect, but misses the adjustment margin that determines long-run protection.
- *Small-country / partial-equilibrium assumption.* The derivation fixes world prices of both the final good and its imported inputs. For a large country, the formula can overstate protection because some of the tariff is absorbed by foreign exporters. General-equilibrium effects are also ignored.
- *Silence on non-tariff barriers.* Effective protection counts only nominal tariffs. Quotas, licensing, domestic content rules, antidumping duties, and procurement preferences all generate real protection but sit outside the formula.

Problem 4

Restatement: Identify the conditions that produce Bhagwati's "immiserizing growth" outcome, name the type of growth most likely to trigger it, and connect the concept to Prebisch's argument about the secular deterioration in the net barter terms of trade. A general-equilibrium graph is required.

The conditions for immiserizing growth

Solution: Immiserizing growth is the counterintuitive case where a country's economic growth—an outward shift of its production possibilities frontier (PPF)—leaves the country *worse off* in welfare terms. The outcome requires the simultaneous presence of four conditions:

1. **Large-country status.** The growing country must be large enough in world markets that its expanded production affects world prices. A small country is a price-taker, so it can never induce a terms-of-trade deterioration by growing.

2. **Export-biased (ultra-pro-trade-biased) growth.** Growth must be disproportionately concentrated in the export sector—expanding the PPF more along the export-good axis than the import-good axis. Import-substituting growth cannot produce immiserization because it contracts, rather than expands, the country's demand on world markets.

3. **Inelastic foreign demand for the export good.** Foreigners' demand for the country's exports must be sufficiently price-inelastic that an outward shift in export supply causes a sharp fall in world price. If foreign demand is highly elastic, expanded supply is absorbed without a terms-of-trade collapse.

4. **Trade dependence.** The country's trade must represent a large share of its economy, so that a deterioration in terms of trade transmits significant income losses to the domestic economy.

The type of growth most likely to trigger immiserization is **export-biased growth**—particularly when it occurs in a primary commodity sector with inelastic world demand. The mechanism: growth expands production of the export good, which (given inelastic foreign demand) forces a disproportionate fall in the world export price; the resulting terms-of-trade loss exceeds the real income gain from the expanded PPF, leaving the consumption bundle on a lower community indifference curve than before growth.

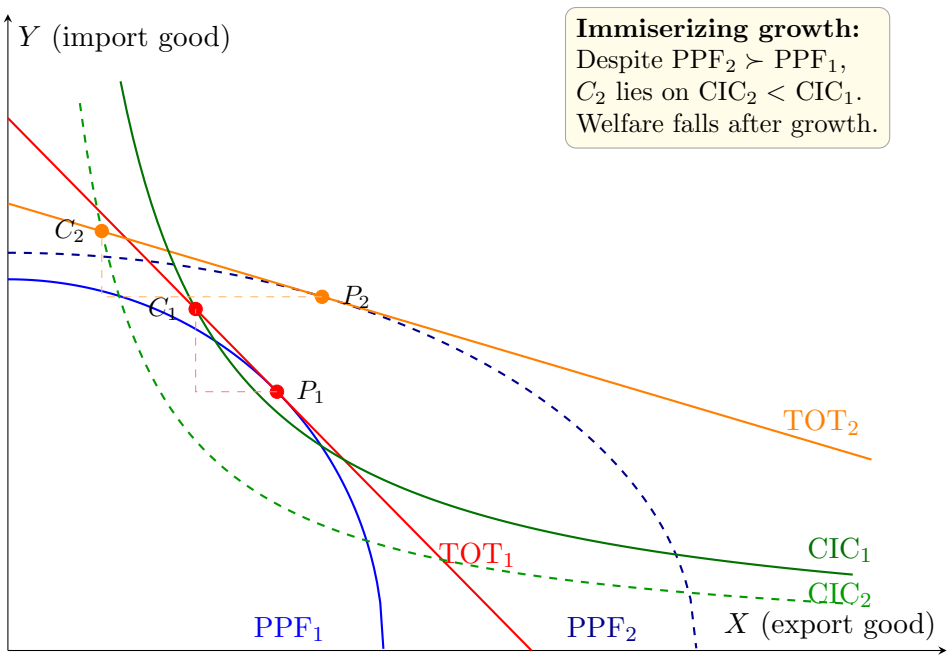

Figure 3: Bhagwati's immiserizing growth. The PPF expands outward in an export-biased way (PPF$_1 \to$ PPF$_2$), but inelastic foreign demand collapses the terms of trade (TOT$_1 \to$ TOT$_2$; slopes $-1.2 \to -0.35$). The new production point P_2 sits on PPF$_2$ at the TOT$_2$ tangency; consumption shifts from C_1 on CIC$_1$ to C_2 on CIC$_2$, which is strictly closer to the origin. The country exports more X and imports more Y in volume, but is worse off in welfare terms.

Connection to Prebisch's terms-of-trade deterioration thesis

Solution: Prebisch's secular-decline hypothesis and Bhagwati's immiserizing-growth theorem are, in a sense, two statements of the same underlying mechanism—one empirical, one theoretical.

Prebisch argued that developing countries specializing in primary-commodity exports face a long-run tendency for their net barter terms of trade to deteriorate against developed countries exporting manufactures. His mechanism rests on three empirical asymmetries:

- **Demand asymmetry.** As world income grows, demand for primary commodities grows slowly (income inelasticity, Engel's Law) while demand for manufactures grows rapidly (income elasticity above unity).
- **Market-structure asymmetry.** Productivity gains in developed-country manufacturing are captured by organized labor and capital as higher wages and profits; productivity gains in developing-country primary-commodity sectors—operating in more competitive markets with weaker labor institutions—are passed on to buyers as lower prices.
- **Supply response asymmetry.** Developing countries, pressed for foreign exchange, cannot reduce export volumes when prices fall; they often expand exports instead, further depressing world prices.

The connection is direct: Prebisch's three asymmetries are exactly the conditions required for Bhagwati's immiserizing growth to operate. A developing country that grows by expanding its primary-export sector (export-biased growth) into inelastic world demand (Engel's Law) is running Bhagwati's model in real time. Bhagwati provided the formal general-equilibrium theory;

Prebisch argued this theory described the actual post-war trajectory of Latin America, Africa, and the commodity-dependent developing world. The policy implication both drew—diversification away from primary exports and, in Prebisch's case, import-substitution industrialization—follows from the same diagnosis: that unqualified pursuit of comparative advantage in inelastic-demand commodities can be welfare-reducing when a country is large enough in its export market to move prices against itself.